

Project Statement: Student Record Management System

1. Problem Statement

Managing student records and academic performance manually can be inconvenient, time-consuming, and prone to human error—especially when performing repetitive calculations like total marks, percentages, and pass/fail determinations. Existing simple workflows often lack a structured system to store student data and compute results quickly.

To solve this, the **Student Record Management System** provides an organized, automated, and Python-based command-line interface (CLI) application. It allows administrators to store student details using modular data structures and automatically perform essential operations such as record entry, searches, grade updates, and performance evaluations.

2. Scope of the Project

The project serves as an in-memory academic management application designed to demonstrate structured programming using Python.

In-Scope:

- Managing student details (Roll Number, Name, Marks for 5 subjects) during runtime.
- Menu-driven command-line interaction.
- Automatic computation of aggregate metrics (Total Marks, Percentage) and final evaluation (PASS/FAIL).
- Dynamic searching and updating of existing student records using dictionary structures.

Out-of-Scope (Future Enhancements):

- Permanent data persistence using file handling or database integration (e.g., SQLite, PostgreSQL).
- Graphical User Interface (GUI) or Web Interface.
- Multi-role authentication (e.g., separate Admin/Teacher/Student portals).
- Support for variable numbers of subjects or custom grading logic.

3. Target Users

- **Academic Instructors & Teachers:** Who need a quick tool to enter subject marks, evaluate student performance, and view summary metrics.
- **School/Department Administrators:** Who require a lightweight local tool to manage student details and perform quick lookups by roll number.
- **Programming Students & Educators:** Who use the system as a foundational reference model to study Python concepts such as dictionaries, lists, functions, loops, and control

structures.

4. High-Level Features

1. **Add Student Record:** Allows input of a unique roll number, student name, and marks for five subjects, preventing duplicate entries.
2. **Search Student:** Quick lookup of an existing student record by roll number to view individual subject scores and details.
3. **Update Marks:** Enables editing and overwriting marks for five subjects of an existing student.
4. **Display Result & Automated Evaluation:** Calculates total marks and percentage, automatically assigning a **PASS** or **FAIL** status based on a predefined passing threshold ($\geq 40\%$).
5. **Interactive Menu Interface:** A continuous, loop-based command-line menu providing seamless navigation and robust feedback for invalid inputs.

MADE BY

NAME - SNEH TIWARI

RNO. 26BCE10069

PROJECT NAME - PYTHON ESSENTIALS